



Carbon emissions are projected to rise by 1.1% by the end of 2025.

WHEN WILL GLOBAL GREENHOUSE-GAS EMISSIONS PEAK?

Scientists say emissions could begin to decline in the next several years.

By Jeff Tollefson

Global fossil-fuel emissions are likely to hit yet another record high this year, scientists announced on 12 November at the COP30 UN climate summit in Belém, Brazil, but there are signs that the world is approaching a peak. Indeed, some researchers argue that carbon dioxide emissions might have already begun to decline, and many think that overall greenhouse-gas emissions (including methane, nitrous oxide and fluorinated gases) could begin to decrease by the end of the decade.

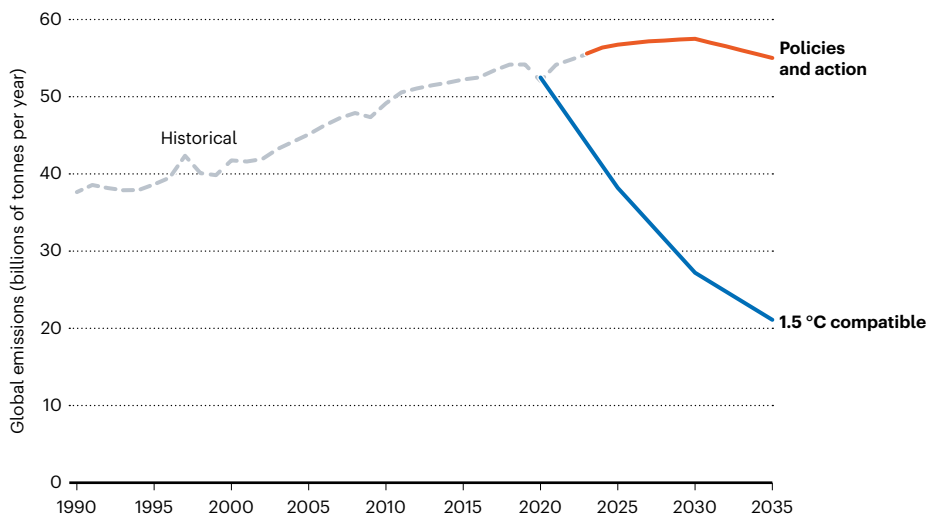
Emissions from the burning of fossil fuels and cement production are projected to rise by 1.1%, to 38.1 billion tonnes of CO₂, this year, according to data published on 13 November by the Global Carbon Project, an international consortium of researchers who track carbon emissions (see go.nature.com/43pm6hv).

Overall carbon emissions could decline slightly if a projected drop in deforestation and other land-use changes is factored in, but researchers warn that it's still too early to say that the world has turned a corner on its fossil-fuel addiction (see 'When will carbon emissions peak?').

"We don't [project] the global inflection point until around 2030, unfortunately, but it does look like emissions are flattening off," says Bill Hare, a physicist who is head of Climate Analytics, a non-profit consultancy in Berlin that analyses the impact of global climate policies.

WHEN WILL CARBON EMISSIONS PEAK?

Greenhouse-gas emissions have been increasing every year since 1990. When emissions will peak, and how much they will decline, depends on governmental policies and actions.



As things stand, greenhouse-gas emissions are around 10% higher than a decade ago, when the Paris climate agreement was signed, and a far cry from what is needed to limit warming to 1.5 °C above pre-industrial levels, as called for under the treaty.

The China factor

Emissions among the major industrialized countries, which are responsible for the bulk of historical emissions, have been falling for more than two decades (see 'Which regions emit the most greenhouse gases?').

However, they are rising nearly everywhere else. Although much of the focus tends to fall on rapidly developing juggernauts such as India and China, the emissions growth is widespread as low- and middle-income countries seek to grow their economies and fulfil energy demands.

By far the biggest driver over the past two decades has been China, which now accounts for nearly one-third of the world's greenhouse-gas emissions. The main culprit is the country's vast and still-growing fleet of coal-fired power plants, which burnt nearly 2.3 billion tonnes of coal last year, according to the International Energy Agency.

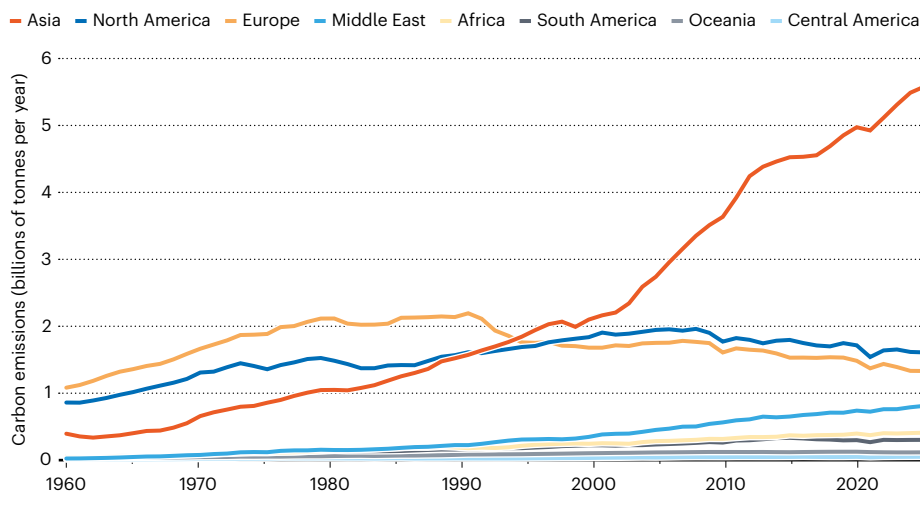
There are signs of change, however, with China now leading the global clean-energy sector. The country is rapidly turning to electric vehicles as well as wind and solar energy and has committed to reducing its overall greenhouse-gas emissions by at least 7% from peak levels by 2035. When China's emissions peak, so, too, will global emissions, predicts Hare.

Have China's emissions peaked?

Others argue that China's carbon emissions have already begun to decline. Data from Carbon Monitor, a frequently updated data set that began tracking daily carbon emissions during the COVID-19 pandemic, suggest that

WHICH REGIONS EMIT THE MOST GREENHOUSE GASES?

The majority of greenhouse-gas emissions come from Asian countries, notably China, which have rapidly expanded their fossil-fuel energy sectors and cement-production capacities. European countries, the United States and Canada have reduced their emissions.



China's carbon emissions peaked in 2024 and will fall by 1.2% this year (see go.nature.com/3zxyvjn). Scientists at the Centre for Research on Energy and Clean Air (CREA), a think tank based in Helsinki, have also documented a downward trend (see go.nature.com/48slxe9).

Although the expansion of electric vehicles and renewable energy is having an impact, the biggest factor in the current decline is the collapse of the Chinese real-estate market, which has resulted in less demand for cement and steel, says Zhu Liu, an Earth-systems scientist at Tsinghua University in Beijing.

Given current energy and economic trends, as well as China's climate goals, Liu says, emissions of CO₂, at least, are likely to continue to decline. "I would say this is the peak of China's carbon emissions," Liu says.

If that's the case, the question is whether emissions of other greenhouse gases will follow.

The peak might end up looking like a series of rolling hills owing to uncertainties and annual variations in energy consumption, says Glen Peters, a climate-policy researcher at the Cicero Center for International Climate Research in Oslo who is involved in the Global Carbon Project. US emissions are projected to go up this year, partly because colder-than-usual winter temperatures increased demand for heating, in stark contrast to the unseasonably warm winters in the two previous years.

"I would want to see emissions going down for several years before I would stand up on a building top and shout that emissions have peaked," Peters says.

Going down

Hitting a peak is also just the first step: as long as the world continues to emit greenhouse gases, the climate will continue to warm. Achieving the goals of the Paris agreement requires reducing global carbon emissions to zero by around

mid-century. And because average global temperatures are almost certain to surpass the 1.5°C goal in the next decade, humanity will also

need to remove CO₂ from the atmosphere to cool down the planet later this century to meet the goals of the Paris agreement.

Already, scientists around the globe are investigating a range of CO₂ removal methods that could be ramped up in the coming decades. In some cases, the focus is on nature-based solutions, such as planting trees or modernizing agricultural systems to build up carbon in soils. In others, scientists are developing industrial systems that could be used to extract carbon from the atmosphere directly.

But it will take time to test, develop and scale up such solutions, and in the meantime global-warming impacts are projected to accumulate.

Climate scientist Katharine Hayhoe compares the management of greenhouse-gas levels in the atmosphere to maintaining a swimming pool. "We have to turn off the hose," says Hayhoe, who serves as chief scientist for the Nature Conservancy, a conservation group based in Arlington, Virginia. "But we also have to make the drain bigger, and we have to learn to swim."

SOURCE: P. FRIEDLINGSTEIN ET AL. EARTH SYST. SCI. DATA 17, 965–1039 (2025)

CRISPR VS CHOLESTEROL: CAN GENE EDITING PREVENT HEART DISEASE?

Results from a small trial highlight a growing interest in developing such therapies for common illnesses.

By Heidi Ledford

A CRISPR–Cas9 gene-editing therapy has halved people's cholesterol levels in a small clinical trial – raising hopes that, with further study, gene editing could one day be harnessed to provide a one-stop treatment for a common cause of heart disease.

For the study, researchers used CRISPR to disable a gene, called *ANGPTL3*, that helps to regulate levels of fatty molecules, including low-density lipoprotein (LDL) or 'bad' cholesterol and triglycerides, in the blood. Both types of molecule are linked to an increased risk of cardiovascular disease, and levels of both fell by roughly 50% in people who were treated with the highest therapy dose (L. J. Laffin *et al.* *N. Engl. J. Med.* <https://doi.org/g99wpr>; 2025).

So far, only 15 people have received the treatment. But, if all goes well in future studies, investigators hope that gene editing might one day stop thousands of people needing to take cholesterol-lowering medications daily.

"What a revolution to be able to do that,"

says Luke Laffin, a specialist in preventive cardiology at the Cleveland Clinic in Ohio and a lead investigator on the trial. "Conceptually, it's a great idea: we can move from chronic therapy to something that's one-and-done."

That revolution will take years: more safety tests would be needed to make the concept a reality, he adds. And there are questions about how much the therapy would cost, and whether consumers would be willing to embrace it. But the trial results are the latest to show promise in harnessing gene-editing techniques to tackle some of the most common diseases.

Decade-long dream

"The dream scenario for me is getting these gene-editing therapies broadly applied," says Kiran Musunuru, a cardiologist at the Perelman School of Medicine at the University of Pennsylvania in Philadelphia.

A little over a decade ago, Musunuru wondered whether CRISPR–Cas9 gene editing could be harnessed to prevent or treat the world's leading killer: cardiovascular disease. He pitched the idea to a few venture capitalists.